Image Analysis and Data Fusion

Image Analysis and Data Fusion (IADF TC)

MISSION

The Image Analysis and Data Fusion Technical Committee (IADF TC) of the Geoscience and Remote Sensing Society serves as a global, multi-disciplinary, network for geospatial image analysis (e.g., **machine learning, deep learning, image and signal processing, and big data**) and data fusion (e.g., **multi-sensor, multi-scale, and multi-temporal data integration**). It aims at connecting people and resources, educating students and professionals, and promoting theoretical advances and best practices in image analysis and data fusion.

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(https://www.grss-ieee.org/wp-content/upload scaled.png)

The Contest: Goals and Organization

Rapid advances in spaceborne Synthetic Aperture Radar (SAR) have transf global, all-weather, day-night monitoring with ever-increasing temporal den intuitive visual detail, it is constrained by cloud cover and illumination. SAR regardless of weather or lighting, making it uniquely suited for tracking sub processes, and rapid environmental events.

Recent advances in commercial SAR satellite systems have transformed h constellations are able to provide sub-meter imagery with flexible tasking a traditional large SAR systems are optimized for repeat-pass geometries an SAR systems prioritize task-based, event-driven acquisitions, leading to gre geometry across temporal stacks.

These temporal stacks may be densely sampled, but typically exhibit subst incidence angle, look direction, and orbital geometry – necessitating an evc problems are approached. While this geometric variability complicates tem opportunities. The increased temporal density and diversity of observation that can be exploited by modern analytical methods.

The 2026 IEEE GRSS Data Fusion Contest, organized by the Image Analysi Committee of GRSS and Capella Space aims to foster the development of i operational challenges encountered in the exploitation of commercial high-

To this end, the contest dataset consists of densely sampled, InSAR compa resolution X-band SAR imagery acquired across a wide range of imaging m the first time, a dense temporal stack of high-resolution InSAR-compatible constellation is being made available at scale, providing a unique testbed fc richness while remaining robust to acquisition diversity.

Participants will submit a short description outlining their method, key insig contest dataset. The aim is to encourage creative use of globally distribute techniques that can advance our ability to interpret and exploit long-term S.

Scientific papers describing the best entries will be included in the Technica presented in an invited session "IEEE GRSS Data Fusion Contest," and publi Proceedings. Furthermore, organizers and contest winners will co-author a (https://www.grss-ieee.org/publications/journal-of-selected-topics-in-applie sensing/) describing the winning approaches in greater detail.

Competition Phases

The contest consists of a single phase in which participants receive the rel develop, explore, and evaluate their methods. Contestants may use the pro analytical or visualization approach they consider relevant. Once their meth short description outlining their idea, methodology, and results obtained wi submission deadline, the organizers will review all entries and select the wi and usefulness of the proposed approaches (see below). After evaluation c announced. Following this, they will receive feedback to write their final ma IGARSS 2026 proceedings. Manuscripts are 4-page IEEE-style formatted (r page limit). Each manuscript describes the addressed problem, the propos results.

Calendar

- February 04: Contest opening: release of data
  The evaluation committee starts to accept submissions.
- April 06: The evaluation committee stops to accept submissions.
- April 13: Winner announcement
- April 20: Internal deadline for papers, DFC Committee review process
- April 28: Submission deadline of final papers to be published in the IG
- August 9-14: Presentation at DFC-dedicated IGARSS 2026 Session

The Data

Dataset Overview

This dataset consists of multiple densely sampled temporal stacks of very Aperture Radar (SAR) imagery collected by the Capella Space satellite cons support advanced research in time-series analysis, change detection, objec machine-learning-based SAR analytics, with a particular emphasis on robu dataset spans a wide range of imaging geometries, acquisition modes, anc real-world operational diversity rather than a synthetic benchmark.

A core motivation behind this dataset design is to encourage the developm algorithms, particularly for participants using deep neural networks (DNNs)

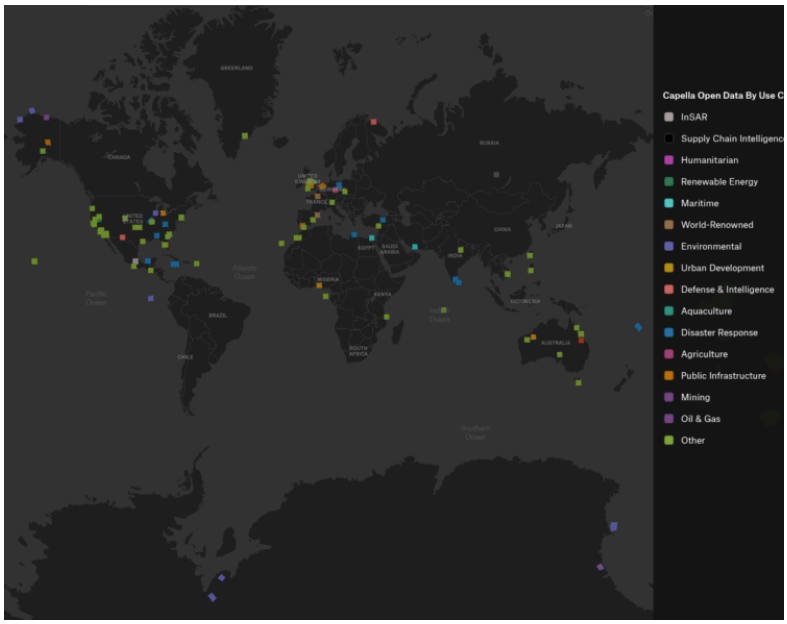
Analysis Possibilities

This dataset supports a broad range of exploratory analyses enabled by de sampling and diverse acquisition geometry. SLC products can be paired to stacks for interferometric change analysis and time-series deformation stu inclusion of long-dwell, high-resolution spotlight-style collections also enab retrieval approaches (e.g., exploiting aperture diversity and geometric prior stacks provide a foundation for pattern-of-life and activity characterization, persistent change, episodic events, and evolving scene dynamics over time

Participants are encouraged to pursue creative uses of the dataset and to i such as land cover, infrastructure, and environmental setting – when devel

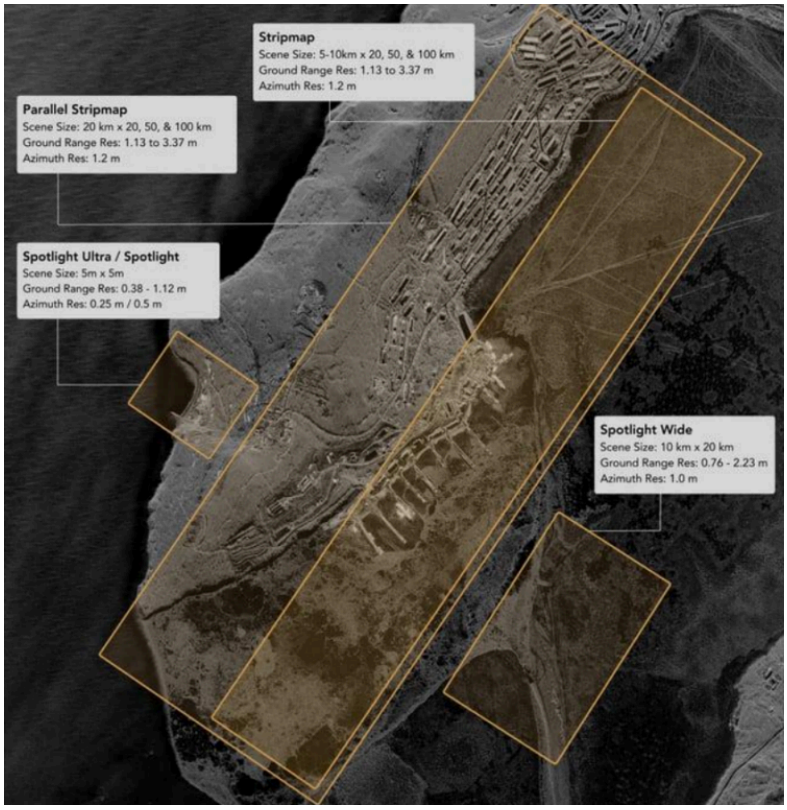
Data Products and Metadata

Approximately 1582 unique collects are provided in coordinated stacks coll of creating 17000+ possible pairs! Stacks over some AOIs are deep with ov course of the year. A geographic distribution of the images are shown in Fi benefits machine-learning approaches by encouraging more generalizable,



content/uploads/2026/02/dfc26-map.png)Figure 1 Geographics distributic
The modes represented in the data set and illustrated in Figure 2 are (more
be found here (https://support.capellaspace.com/what-are-capellas-collect

- **Spotlight mode** provides the highest spatial resolution by continuousl fixed ground location, enabling detailed characterization of fine-scale s extent.
- **Sliding Spotlight** (Spotlight Wide) mode extends scene coverage by g track during collection, offering a balance between high spatial resolut
- **Stripmap** mode images the surface with a fixed antenna pointing as tl producing wider-area coverage with consistent geometry and modera large-scale monitoring.



content/uploads/2026/02/capella.png)Figure 2 Standard imaging mode or

The accompanying CSV file (https://www.grss-ieee.org/wp-content/uploads/2026/02/Capella_IEEE_DataContest_2026.csv) contains r
are as follows:

- **collect_id_ref**: Unique identifier for the primary (reference) SAR acquis
comparison.
- **collect_id_sec**: Unique identifier for the matched secondary SAR acqu
reference collect.
- **start_time_ref**: Acquisition start time of the reference collect, express
- **start_time_sec**: Acquisition start time of the secondary collect, expres
- **platform_ref**: Identifier of the satellite platform that acquired the refer
- **platform_sec**: Identifier of the satellite platform that acquired the sec
- **polarization_ref**: Transmit–receive polarization of the reference collec
- **polarization_sec**: Transmit–receive polarization of the secondary coll
- **mode**: SAR imaging mode used for the acquisition (e.g., Spotlight, Slid
- **look_direction**: Side-looking geometry of the acquisition relative to the
looking).
- **flight_direction**: Orbit direction during acquisition (ascending or desce
- **incidence_ref**: Incidence angle of the reference collect at scene center
- **incidence_sec**: Incidence angle of the secondary collect at scene cent
- **dif_incidence**: Absolute difference in incidence angle between the refe
expressed in degrees.
- **delta_graze_deg**: Difference in grazing angle between the reference ar
degrees.
- **latitude**: Latitude of the scene center or reference point, expressed in
- **longitude**: Longitude of the scene center or reference point, expressec
- **height**: Height above the reference ellipsoid or mean sea level at the s

The dataset includes multiple SAR data product types to support a range o

- **Single Look Complex (SLC)** products provide complex-valued SAR im
phase information, enabling interferometric processing techniques.
- **Geocoded (GEO)** products are image-domain representations projecte
system, facilitating visualization, mapping, and integration with other
- **Complex Phase History Data (CPHD)** products are provided for select
history measurements prior to image formation and enabling custom
and experimental processing.
- **Geocoded Ellipsoid Corrected (GEC)** products provide geocoded ima
to help exclude terrain artifacts.

Accessing the Data

The IEEE Data Fusion Contest 2026 dataset, provided by Capella Space, is
public AWS S3 bucket. Users can access the collection via the published S
(https://stacindex.org/catalogs/capella-space-open-data#/6QUH7kZBffZNEnYuZuzZWv8D3vWA4TEeZvXZ5pubg4QY1ZAgGDY
ip=10&t=2), browse it interactively through a provided felt map (https://felt.
bB24xsH3SuiUlpMdDbVRaA?loc=0,-20.53,1.94z), or retrieve the data progra
datasets or selected assets directly from S3 without authentication.

Submission and Evaluation

Participants submit a concise document describing their contribution. The
overarching idea or goal, the methods or techniques applied, and the result
dataset. Visualizations, examples, and figures should be included to illustra

insights. Other material (animations, additional figures, videos, etc.) can be
Submissions should focus on how the proposed approach leverages the la
what value it brings to understanding, exploring, or analyzing such datasets

The document should use the IGARSS 2026 fullpaper template (https://2026.ieeeigarss.org/papers/paper_kit.php). It must contain the nar
all team members. Adding team members after submission is not possible
one team.

All submissions will be reviewed by a committee. Each contribution will be
criteria designed to reward technical rigor, creativity, clarity, and practical re
the winning entries based on an overall evaluation across the following dim

- **Soundness and technical correctness:** The method should be scienti
correctly implemented. Assumptions must be reasonable, and conclu
- **Originality and creativity:** The approach should introduce novel ideas,
uses of the data. Creative problem framing or unconventional explorat
- **Insightfulness:** Submissions should demonstrate meaningful underst
perspectives, patterns, or phenomena. Insight can arise from methods
- **Usefulness and practical relevance:** The method or findings should p
analysis workflows, enabling new applications, or offering tools for int
- **Effective use of the dataset:** The approach should leverage the chara
series. Methods that make strong use of temporal information, spatia
viewed favorably.
- **Computational efficiency or scalability:** Since the dataset is large-sca
scalability could be included, though this is optional depending on cor
- **Clarity and quality of presentation:** The submission should be well-st
understand. Claims must be supported by evidence or results.
- **Quality of visuals and illustrations:** Figures, plots, or visualizations sh
and helpful for understanding the proposed approach or results.
- **Reproducibility / transparency:** Participants are required to make thei
GitHub). The publicly shared implementation should be sufficiently cle
described method understandable and potentially replicable, thereby s
value.

Resources and Tutorials

Tutorials

- Access and Download (https://stacindex.org/catalogs/capella-space-
data#/6QUH7kZBffZNEnYuZuzZWv8D3vWA4TEeZvXZ5pubg4QY1ZA
ip=10&t=2)
- Python notebook with examples on how to query and filter a STAC col
ieee.org/wp-content/uploads/2026/02/STAC_to_CSV.ipynb_.zip)
- Scaling GEO Images in QGIS (https://support.capellaspace.com/hc/en
Scaling-GEO-Images-in-QGIS) by Capella Space (https://www.capellas
- Opening and Exploring the Geo Images (https://www.grss-ieee.org/co
committees/opening-and-exploring-the-geo-images/)

Tools & Applications

- Python SDK for api.capellaspace.com (https://capella-console-client.r
Space (https://www.capellaspace.com/)
- Single Look Complex data reader for Capella SLC images – python m
into an amplitude image. (https://github.com/capellaspace/capella-re
(https://www.capellaspace.com/)

Publications

- Analyzing LiDAR and SAR data with Capella Space and TileDB (https:// sar-data-with-capella-space-and-tiledb) by Stavros Papadopoulos
- Open SAR data and scalable analytics (https://medium.com/tiledb/op bfd1a5257e9) by Norman Barker
- Radar Generalized Image Quality Equation Applied to Capella Open Da (https://ieeexplore.ieee.org/document/9764201/) by Wade Schwartzk Farquharson, Craig Stringham, Michael Duersch, Jordan Heemskerk

Results, Awards, and Prizes

The first to fourth-ranked teams will be recognized as winners. To be eligibl contribute to the community by sharing their code openly, such as on GitHu

- Present their approaches in a dedicated DFC26 session at IGARSS 20
- Publish their manuscripts in the Proceedings of IGARSS 2026
- Receive IEEE Certificates of Recognition
- Be awarded during IGARSS 2026, Washington, USA in August 2026
- Co-author a journal paper summarizing the DFC26 outcomes, which v IEEE JSTARS. The costs for open-access publication will be supporte

The Rules of the Game

- The dataset can be openly downloaded from the Capella Space Open
- To enter the contest, participants must read and accept the Contest T
- The results should be submitted as a pdf via the corresponding subm (https://forms.gle/gnmu1fKJAqNcxbK98). Optional supplemental mat animations, etc.) can be submitted by providing download links within submission page is based on Google forms. Should the page not be a email to dfc26@googlegroups.com (mailto:dfc26@googlegroups.com in the mail body: Primary Contact Name (First Name Last Name), Prin (Name of University or Organization), Primary contact's email, Country (Optional), Team Members List (List all team members including their (Format: Name – Affiliation – Email)), Link to Source Code (e.g., GitHu this team is participating in another team for this year DFC, Confirmat (e.g., GitHub)).
- Ranking between the participants will be based on the criteria as desc Evaluation Section.
- The committee accepts submissions from February 04, 2026 until Ap
- Each team needs to submit a short paper of max. 4 pages (excl. refere team members, describing the used approach, and showing results. T 2026 paper template and should be submitted via the submission pag (https://forms.gle/gnmu1fKJAqNcxbK98) (see above).
- For the winning teams, the internal deadline for full paper submission paper submission deadline is April 28, 2026.
- Important: Only team members explicitly stated on these documents of the DFC, i.e., being eligible to be awarded as winners and joining the potential publications (IGARSS26 and JSTARS articles). Furthermore, i.e., one person can only be a member of one team. Adding more team description for evaluation is not possible.
- Persons directly involved in the organization of the contest, i.e., the (co organizers are not allowed to enter the contest. Please note that IADF contest. They have been excluded from relevant information concerni fair competition.

Failure to follow any of these rules will automatically make the submission not being evaluated and disqualification from the contest.

Participants in the Contest are requested not to submit an extended abstra
DFC to IGARSS 2026 by the corresponding conference deadline. Only conte
corresponding to the four best-ranking submissions) will submit a 4-page p
Contest by April 20, 2026. The received manuscripts will be reviewed by the
and reviews sent to the winners. Winners will submit the 4-pages full paper
Contest by April 28, who will then take care of the submission to the IGARS
Contributed Session, for inclusion in the IGARSS Technical Program and Pr

For any questions, please contact the organizers at dfc26@googlegroups.c
(mailto:dfc26@googlegroups.com).

Acknowledgments

The IADF TC chairs would like to thank Capella Space for providing the data
supporting the annual Data Fusion Contest.

(https://www.grss-ieee.org/wp-content/uploads/2026/02/Capella-GRSS-IA

More Technical Committees

(https://www.grss-ieee.org/technical-committees/quantum-earth-science-and-technology-quest/)
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(https://www.grss-ieee.org/technical-committees/remote-sensing-environment-analysis-and-climate-technologies-technical-committee-react-tc/)
Remote sensing Environment, Analysis and Climate Technologies